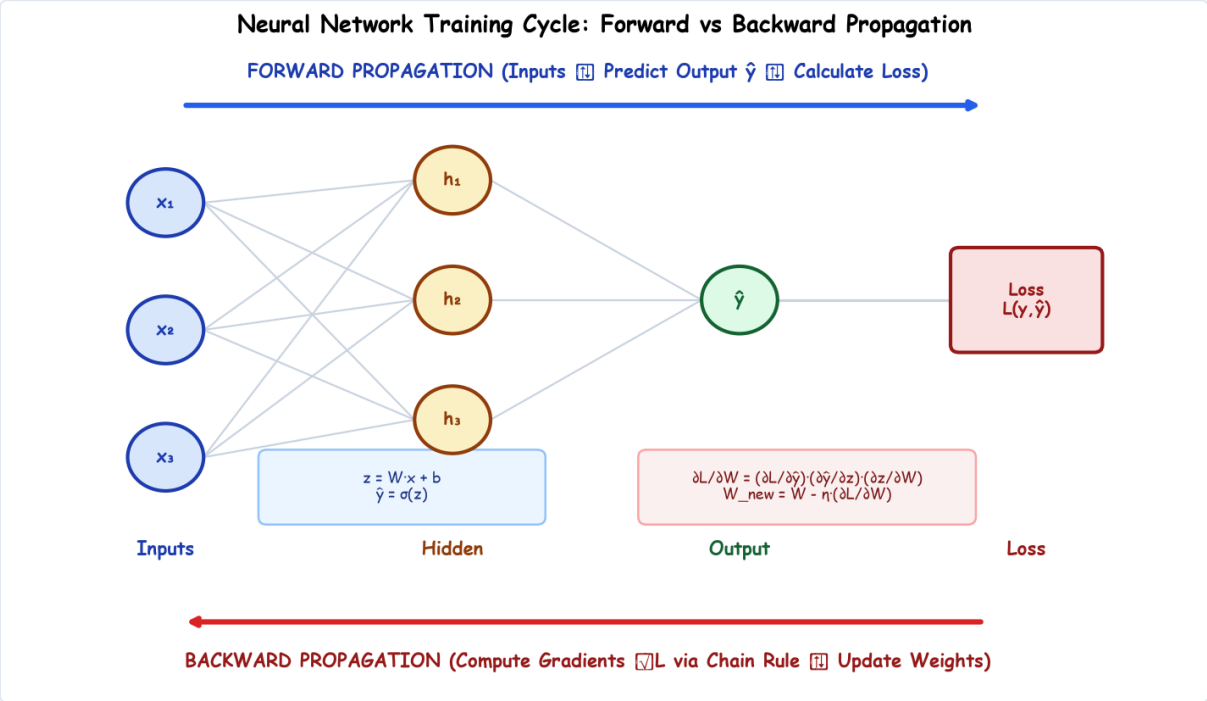
Lecture 37: Deep Learning Foundations (Part-2) — Training Neural Networks

Welcome to the comprehensive master curriculum for **Lecture 37: Deep Learning Foundations (Part-2)**. While Lecture 36 established the architectural representation of artificial neurons and multi-layer perceptrons, Lecture 37 is dedicated entirely to **how neural networks learn**: loss function formulation, backpropagation calculus, gradient descent dynamics, activation selection, and modern adaptive optimizers.

High-Level Training Dynamics: Forward vs Backward Pass

Neural network training is a continuous feedback loop between inference and optimization:



The Governing Mathematical Equations

1. **Forward Pass (Affine + Activation)**:

$$z^{[l]} = W^{[l]}a^{[l-1]} + b^{[l]}, \quad a^{[l]} = g^{[l]}(z^{[l]})$$

2. **Loss Evaluation**:

$$\mathcal{L}(y, \hat{y})$$

3. **Backward Pass (Chain Rule Error Propagation)**:

$$\delta^{[l]} = ((W^{[l+1]})^T \delta^{[l+1]}) \odot g^{[l]'}(z^{[l]})$$

4. **Parameter Updation (Gradient Descent Step)**:

$$W^{[l]} \leftarrow W^{[l]} - \eta \frac{\partial \mathcal{L}}{\partial W^{[l]}}, \quad b^{[l]} \leftarrow b^{[l]} - \eta \frac{\partial \mathcal{L}}{\partial b^{[l]}}$$

Comprehensive Submodule Roadmap

This curriculum comprises 10 exhaustive submodules accompanied by an industrial practice suite. Every module features dual-language documentation (English `README.md` and Hinglish `README_HINGLISH.md`), notebook-style hand-drawn educational diagrams, mathematical proofs, and compiled vector PDFs:

Submodule	Topic Title	Core Concepts & Formulas Covered	Educational Diagram	Documentation Links
37.01	**Forward & Backward Propagation**	Dual-engine training loop, linear combiner, activation, loss, credit assignment	Diagram	English | Hinglish
37.02	**Loss Functions (Regression)**	Loss vs Cost, MSE (L_2), MAE (L_1), Huber Loss, outlier sensitivity analysis	Diagram	English | Hinglish
37.03	**Loss Functions (Classification)**	Binary Cross-Entropy (Log Loss), Softmax layer, Multi-class Categorical Cross-Entropy	Diagram	English | Hinglish
37.04	**Weight & Bias Updation**	Gradient descent update rules, learning rate η dynamics, positive/negative slope mechanics	Diagram	English | Hinglish

Submodule	Topic Title	Core Concepts & Formulas Covered	Educational Diagram	Documentation Links
37.05	**Chain Rule of Derivatives in NN**	Computational graph, local Jacobian evaluation, reverse gradient accumulation	Diagram	English | Hinglish
37.06	**Vanishing Gradient Problem**	Sigmoid derivative upper bound $\sigma' \leq 0.25$, exponential gradient decay $(0.25)^L$	Diagram	English | Hinglish
37.07	**ReLU & its Variants**	Standard ReLU, Dying ReLU pathology, Leaky ReLU, Parametric ReLU (PReLU), ELU	Diagram	English | Hinglish
37.08	**Batch vs Iteration vs Epoch**	Definitions, governing formula $\text{Steps} = N/B$, hardware throughput trade-offs	Diagram	English | Hinglish
37.09	**Classical Optimizers**	Batch Gradient Descent (BGD), Stochastic GD (SGD), Mini-Batch GD, loss surface trajectories	Diagram	English | Hinglish
37.10	**Modern Variants of GD**	Momentum, RMSProp, Adam (1st & 2nd moments + bias correction), industrial standards	Diagram	English | Hinglish

| **Practice** | **Industrial Practice Suite** | 15 advanced mathematical, analytical, and implementation problems with complete solutions | N/A | Questions | Solutions |\n